Cross-Environment Human Activity Recognition from WiFi CSI: A Controlled Study of Adversarial and Contrastive Training Objectives

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***Abstract*—WiFi Channel State Information (CSI) enables device-free human activity recognition, but a model trained in one room degrades when it is moved to another, and cross-environment generalization is the main obstacle to deployment. We ask whether two popular representation-shaping objectives — an adversarial domain classifier trained through a gradient reversal layer (GRL) and a supervised contrastive (SupCon) loss — improve transfer over plain cross-entropy (CE), under conditions strict enough for the answer to mean something: one backbone, one data pipeline, one seed and one evaluation harness on the SHARP micro-Doppler dataset under a leave-one-environment-out protocol, with checkpoints selected on validation only, the test set evaluated exactly once, and differences assessed by a trace-level paired bootstrap. Neither objective helps, and each fails for its own measured reason. Our deep residual CE model reaches macro-F1 0.804 on the unseen environment. SupCon loses 7.8 accuracy points with a bootstrap interval that excludes zero, and the deficit is in the representation, not in the readout: kNN, nearest-class-mean and a full fine-tune all fail to recover it, so its features carry less of what separates the activities than CE's do. The GRL loses 15.8 points — more, but unresolved, its damage falling on a few traces — with a $10\times$ larger seed spread, because it has no target: no domain attribute is linearly readable from the features an adversary reads, and the dataset's environment incidence leaves the target degenerate. The reference design's shallow–wide network loses 23.5 points in the identical recipe. We therefore recommend probing a contrastive encoder under several readouts, and the nuisance factor for readability, before investing in either objective.**

***Index Terms*—WiFi sensing, Channel State Information, Human Activity Recognition, Cross-Environment Generalization, Contrastive Learning, Adversarial Training.**

I. Introduction

Wireless sensing turns existing WiFi infrastructure into a device-free perception system: a moving person perturbs the multipath propagation of the radio signal, and those perturbations are recorded in the Channel State Information (CSI) that commodity network cards expose. Human Activity Recognition (HAR) from CSI needs no wearable, no camera and no cooperation from the subject, which makes it attractive for assisted living, surveillance and smart-home applications [1]. The micro-Doppler signature — the distribution of reflected power over velocity and time — is among the most informative representations derived from CSI, and is our input throughout.

What separates CSI HAR from deployment is not in-distribution accuracy, which is by now high, but *generalization across environments*: a model trained in one room entangles the activity with that room's static geometry. The literature answers with two representation-shaping objectives — *adversarial domain invariance*, where a domain classifier behind a gradient reversal layer (GRL) [2] pushes the encoder to discard environment-specific cues, and *supervised contrastive* (SupCon) learning [3], which promises a more transferable embedding geometry. Both are usually reported on the benchmarks where they helped, each with its own backbone and protocol; whether they help on a given dataset, and what must be true of it for them to help, is assumed more often than tested.

We test exactly that, on the SHARP micro-Doppler dataset [4] under a leave-one-environment-out (LOEO) protocol. Front-end, windowing, backbone, optimizer, seed and evaluation harness are held identical, so the objective is the only free variable — batching, two views and view strength follow from the contrastive loss, not chosen; checkpoints are selected on validation alone, the test set is evaluated exactly once from a frozen row list, and differences are assessed with a paired bootstrap that treats the recording, not the window, as the statistical unit. The answer is a controlled negative result, and its value lies in two explanations of equal weight, both obtained by probing the frozen encoders: the contrastive features are not informative enough to separate the activities under any readout we tried, and no domain attribute is linearly readable from the features an adversary would consume, so on data with this structure the adversarial route could not have worked. Both preconditions cost one probe on cached features, and we argue they should be measured before either objective is trained.

Contributions. The key contributions of this report are:

- A controlled LOEO comparison of CE, CE+GRL and SupCon for cross-environment CSI HAR under one apparatus and a single-use test protocol; our CE residual network reaches macro-F1 0.804 on the unseen environment and no variant improves on it.
- Evidence that SupCon transfers *worse* — 7.8 accuracy

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points, the one primary comparison the bootstrap resolves — with the reason measured: non-linear readouts, a concatenation control and a full fine-tune agree that its features are not informative enough to tell the activities apart, so the loss is in the representation and no head can repair it.
- The complementary explanation for the GRL, which loses 15.8 points (unresolved: the damage falls on a few traces) with a $10\times$ larger seed spread: no domain attribute is linearly readable from train features — replicated on five encoders and two rotations — so the adversary has nothing to remove, and the environment incidence makes this structural, not accidental.
- A statistical protocol for small LOEO test sets, under which the backbone alone is worth 23.5 accuracy points — more than any objective we tested.

II. Related Work

WiFi CSI human activity recognition. Early CSI HAR pipelines fed hand-crafted amplitude/phase statistics to shallow classifiers; the field then moved to deep models on time–frequency representations, of which the micro-Doppler spectrogram is widely used [1]. Our reference point is SHARP [4]: it sanitizes the phase of commodity IEEE 802.11 CSI, reconstructs the channel, extracts a Doppler representation and classifies with a shallow, wide convolutional network followed by decision fusion across antennas. SHARP explicitly targets environment- and person-independence, and its own evaluation already shows accuracy degrading on unseen rooms — but it reports that degradation on a single protocol, without isolating what causes it. We adopt the cross-environment setting, tighten it into a strict LOEO protocol with a single-use test session, and reimplement the SHARP network both as a protocol anchor and as a backbone ablation inside our own recipe, separating the architecture axis from the objective axis.

Domain generalization by adversarial invariance. The dominant recipe for making a representation domain-invariant is adversarial: a domain discriminator is trained on the encoder features while a gradient reversal layer (GRL) [2] inverts its gradient into the encoder, so the encoder is optimized to make the domain *unpredictable* while the task head still classifies. WiFi sensing pursues the same invariance [5], adversarially or by aligning feature distributions across training environments, on the implicit and rarely tested precondition that the nuisance factor is present and readable in the features; when it is not, the adversary reduces to a gradient of noise. A central point of this report is to *measure* that precondition.

Contrastive representation learning. SupCon [3] extends the self-supervised contrastive objective of SimCLR [6] by treating all same-label samples as positives, and is reported to yield representations more robust and transferable than cross-entropy ones. The promise is geometric — a class-collapsed, well-separated embedding — but whether that geometry survives a readout, and whether it transfers across environments, is empirical: we interrogate it under several readouts and a full fine-tune, so that a negative result cannot be blamed on one. Other lines pursue generalization by meta-learning [7] or self-supervised pre-training [8]; these are out of scope here, since each replaces backbone and protocol as well as objective — precisely the confound this study avoids.

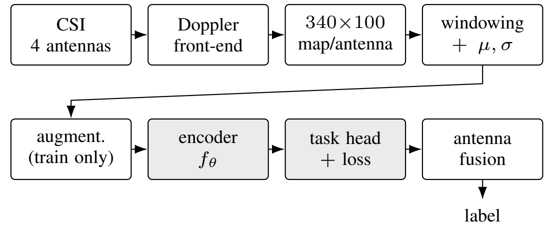

Fig. 1: Processing pipeline. Split and (μ, σ) are frozen before training; the prediction is the argmax of the softmax averaged over the four antennas, worth 8.5 test points over the best single antenna.

III. Processing Pipeline

Fig. 1 shows the path from a raw CSI recording to an activity label, from the Doppler front-end to the fusion of the four antenna scores. Learning acts on the shaded blocks only, and every configuration reuses the same front-end, windowing, encoder, fusion rule and harness, so a difference between two rows is attributable to the training objective and not to incidental engineering. **One configuration is deliberately outside this rule**: C0, the reimplementation of the reference SHARP design, runs on its own split protocol, five-class label set, backbone and majority-vote fusion, anchoring our harness against the prior approach rather than entering the objective comparison (Section VI-B).

IV. Signals and Features

Measurement setup. The SHARP micro-Doppler dataset [4] was collected on a commodity IEEE 802.11 link with *identical hardware throughout*: domains differ by room, monitor position, subject, day and line-of-sight condition, never by radio chipset. It holds three environments (bedroom, living room, laboratory), three subjects and seven *capture sets* — one (room, monitor position, subject, propagation) configuration each — over twelve campaigns; the unit of recording is a *trace*, acquired simultaneously on four antennas, and our inventory counts 101 traces and 404 per-antenna streams. Activities carry the dataset's letter codes: the five-class core is E (empty room), W (walking), R (running), J (jumping) and L (sitting still), and the extended set adds C (sit-down and stand-up), H (arm exercise) and S (standing still).

Front-end, windowing and normalization. The data is distributed as Doppler traces already produced by the reference front-end (CSI phase sanitization, channel reconstruction, and a short-time Fourier transform along the packet axis with hop $T_c \approx 6\,\mathrm{ms}$), giving per antenna a map of reflected power over 100 velocity bins and time, which we consume as-is. Each map is cut into windows of 340 time steps $\times$ 100

velocity bins ($\approx$ 2.0 s), with training stride 100 ($\approx$ 71% overlap) and validation/test stride 340 so that evaluation units are *disjoint*: the primary rotation trains on $\approx$ 53k (window, antenna) samples and is tested on 425 fused windows. Two global scalars (μ, σ), computed on training windows only, are stored in the frozen split file and reused unchanged elsewhere. Both axes carry a physical orientation, so flipping velocity (it inverts the Doppler sign) and reversing time (it maps sit-down onto stand-up) are forbidden augmentations.

Augmentation. On the fly, after standardization, in a fixed order: circular time shift (± 10 frames), time masking (1–2 masks of 5–20 frames), velocity masking (1–2 masks of 2–10 bins) and amplitude scaling ($s \sim \mathcal{U}(0.8, 1.2)$), masks filled with the post-standardization mean. The cross-entropy configurations apply each with probability 0.5; the contrastive one draws *two independent views* of each window at 0.8, with additive Gaussian noise ($\sigma = 0.05$) at 0.5, since SupCon needs views hard enough for its positives to be informative. Velocity is masked sparingly, as that axis carries the feature separating walking from running.

Splits and rotations. Two rules are enforced by blocking assertions: *we split by trace, never by window* — all windows of a trace, on all four antennas, fall on one side, since otherwise the static channel leaks and inflates every metric — and splits are frozen once as JSON under version control. The primary rotation holds out the laboratory (81/9/11 traces), the hardest choice available, and a second holds out the living room (80/6/15) for replication; a held-out capture set varies room, monitor position, subject and day at once, so we speak of held-out *sessions* throughout. Because the class distribution is very uneven across capture sets, every (capture set, activity) cell with fewer than four traces has one trace pinned into training; the price is that no C, E or S trace reaches validation on the primary rotation, making validation macro-F1 a five-class within-run selection signal — common-mode across configurations, and with a per-class effect that Section VI-F measures. A third, leave-the-bedroom-out rotation was rejected before any run: the bedroom holds five of the seven capture sets, so its training side would be two single-session environments totalling 26 traces, with a two-trace validation split — confounding "the model does not generalize" with "the model had no data". That skewed incidence also has a structural consequence (Section VI-E).

V. Learning Framework

Fig. 2 shows how the shared encoder is trained under the three objectives.

Backbones. C1–C4 share a residual convolutional encoder [9] adapted to a 340×100 input whose axes are not interchangeable. A stock ResNet-18 stem would crush the velocity axis to a few bins; instead the stem (conv 3×3 and max-pool, both stride $(2, 1)$) and stages 3–4 downsample *time* only, while one isotropic stride $(2, 2)$ in stage 2 halves velocity once, leaving a final map of $\approx 11 \times 50$. The four residual stages are halved in width (32/64/128/256 channels) and global average pooling yields the feature vector, $d_{\mathrm{enc}} = 256$;

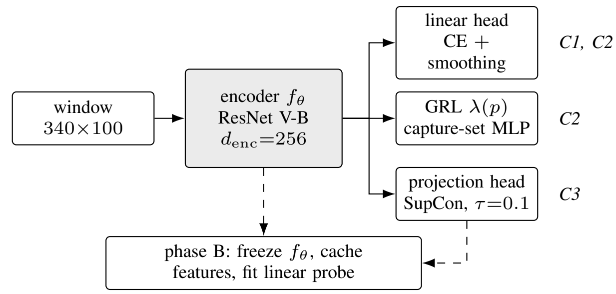

Fig. 2: Learning framework: one encoder, three objectives, tagged at right. The dashed path doubles as the diagnostic instrument: any encoder can be frozen, cached and probed for the activity or for a domain attribute.

at 4.66 GFLOPs per window and 2.79 M parameters, the encoder was selected on a throughput and memory gate, not on accuracy. C0 and the backbone ablation use instead a faithful reimplementation of the SHARP classifier [4]: three parallel shallow branches concatenated, reduced by a 1×1 convolution, flattened to 25 500 features and mapped to the classes by one dense layer — $\approx$ 1k convolutional against $\approx$ 204k dense parameters, a *near-linear wide* model, the opposite design point.

Heads and objectives. All heads are parametrized on d_{enc}. The activity head is linear, trained with cross-entropy and label smoothing 0.1. The adversarial head is an MLP ($d_{\mathrm{enc}} \to d_{\mathrm{enc}}/2 \to 6$ capture sets, dropout 0.3) behind a gradient reversal layer, giving the C2 objective

$$\mathcal{L} = \mathcal{L}_{\mathrm{act}} + \lambda(p)\,\mathcal{L}_{\mathrm{dom}}, \qquad \lambda(p) = \frac{2}{1 + e^{-10p}} - 1, \quad (1)$$

with $p = \min(\mathrm{epoch}/20, 1)$, a ramp fixed in advance so the adversary reaches full strength independently of when early stopping fires. The contrastive head is an MLP $d_{\mathrm{enc}} \to d_{\mathrm{enc}} \to 128$ with ℓ_2-normalized output, trained with the supervised contrastive loss [3] at temperature $\tau = 0.1$. C3 trains in two phases: phase A optimizes it for a fixed horizon of 60 epochs with no in-loop selection, checkpointing on a pre-committed grid (epochs 40/50/60); phase B freezes the encoder, discards the projection head and fits a linear probe on cached features, and only the grid checkpoint with the best validation macro-F1 reaches test. That frozen-probe recipe (Adam, 10^{-3}, ≤ 30 epochs, early stopping on the shared metric) is reused for every probe in this report, so encoders are always compared under one readout.

Configurations. The five configurations span a 2×2 design over loss family and adversary, plus the anchor: C0 (prior approach), C1 (CE, our model and internal control), C2 (CE+GRL), C3 (SupCon) and C4 (SupCon+GRL). C4 was *closed on evidence and never trained*: the diagnostics of Section VI-E show no readable domain target under either loss family, so its expected outcome was "C3 plus noise". Three pre-registered arms complete the picture: C3-ft (a full CE fine-tune from the SupCon encoder), C1-aug (amplitude augmentation) and C1-sharplike (the backbone ablation).

Batching, optimization and reproducibility. The CE configurations draw uniform batches of 256 (window, antenna) samples and optimize an *unweighted* loss while being selected on macro-F1, a declared asymmetry kept for comparability with the reference recipe. The contrastive loss is intra-batch and cannot use uniform batches: a $P \times K$ sampler ($P = 8$ classes, $K = 32$ traces each) draws each class's traces round-robin over the capture sets where the class exists, making positives "same activity, different domain" by construction, and admits at most one window per trace per batch (falling back to a ≥ 340-frame offset for classes with fewer than K traces), since overlapping windows of one recording would reintroduce trivial positives. Optimization is AdamW (weight decay 10^{-4}, peak learning rate 10^{-3}, cosine decay, 5 warm-up epochs), mixed precision, gradient clipping at 1.0, epochs of 400 fixed steps (300 in phase A), at most 40 CE epochs, early stopping on fused validation macro-F1 with patience 10. Seed 42 governs initialization, sampler reseeding and augmentation; two pre-registered seed-43 replicates measure the training-variance floor.

Evaluation harness and the single-use test rule. One shared harness maps a checkpoint to per-set metric files — accuracy, macro-F1, per-class F1, confusion matrices and expected calibration error — always reported with the number of test traces, since that is the effective sample size. Checkpoints are selected on validation only and the *test set is evaluated exactly once*: one session over a frozen, pre-registered row list logged before any test data is read, so evaluate-then-decide selection is impossible.

VI. Results

Statistical protocol. With 11 test traces the *trace*, not the window, is the unit of analysis: the windows of one recording share a channel and are not independent evidence. Differences are assessed by a trace-level *paired* bootstrap ($N = 10^4$) on *accuracy*; macro-F1 is descriptive only, since six of the eight test classes are a single trace and just $\sim$2.8% of resamples contain all eight. Bootstrap and seed variance are never merged, and since $\sim$10 comparisons share the same traces, C1–C2 and C1–C3 are primary and the rest exploratory.

A. Main results

Tab. 1 reports the main configurations with their paired accuracy difference against our model C1. C1 transfers best among them (accuracy 0.805 against a 0.13 majority baseline, seed-stable to 0.5 points) and no variant resolves an improvement over it: SupCon falls 7.8 and the GRL 15.8 points below it, while the largest single effect in the table is not an objective but the backbone, worth 23.5 points at equal recipe.

B. Our model vs. the reference design

C0 is an *anchor*, not a benchmark reproduction: it follows the published architecture and majority-vote fusion on the paper's five-class core, but holds out 20% of the training capture set for early stopping, which the reference implementation does not. It supports no "we reproduced X%" claim, only that our apparatus reproduces the prior approach's *behaviour*, degrading to macro-F1 0.605 over the held-out capture sets as the original work reports. Since C0 differs from our model in protocol, class set, fusion *and* architecture at once, the like-for-like comparison is the ablation *C1-sharplike*: the reference network inside the exact C1 recipe, only the backbone changed. It loses 23.5 accuracy points (test macro-F1 0.543 vs. 0.804), CI $[-0.399, -0.092]$ — "deep beats near-linear-wide in this recipe", on one axis only. Its role is to close the objection that our CE baseline is a strawman, so that the nulls below are measured against a strong control.

TABLE 1: Main results: validation macro-F1 (checkpoint selection; five-class, not comparable to the test columns), test macro-F1 and accuracy on the held-out session, and the paired accuracy difference vs. C1 with its 95% trace-level bootstrap CI ($*$ = interval excludes zero). Top: primary rotation (leave-lab-out, 8 classes, 11 traces). Bottom: second rotation (15 traces) and the 5-class SHARP anchor on its own protocol (57 traces), neither comparable to the top block.

Config	Objective	Val F1	Test F1	Test acc	Δacc vs C1 [95% CI]
C1	CE (our model)	0.887	**0.804**	**0.805**	—
C1′ (s43)	CE, seed 43	0.878	0.799	0.800	-0.005
C2	CE + GRL	0.842	0.601	0.647	-0.158 $[-0.323, +0.034]$
C2′ (s43)	CE + GRL, s43	0.787	0.562	0.600	-0.205
C3	SupCon (probe)	0.819	0.729	0.727	-0.078 $[-0.128, -0.029]^*$
C3-ft	SupCon + CE ft	0.818	0.706	0.720	-0.085 $[-0.221, +0.058]$
C1-sharplike	CE, SHARP net	0.738	0.543	0.569	-0.235 $[-0.399, -0.092]^*$
C1-s6out	CE, living-out	0.776	0.684	0.743	(own rotation)
C0	SHARP anchor (5-cls)	0.892	0.605	0.612	(own split)

C. Supervised contrastive learning does not transfer better

This is the comparison the study was primarily designed to answer, and the only one the bootstrap resolves. C3's contrastive phase converged cleanly — its pre-committed checkpoint grid spans 0.7 validation points (0.819/0.815/0.812 at epochs 40/50/60), so the selected checkpoint sits on a plateau — and on test it lands 7.8 accuracy points below C1, CI $[-0.128, -0.029]$.

Since a frozen linear probe could in principle understate a non-linear representation, we attacked that explanation from four directions. *Readouts:* on the same frozen validation features C3 scores 0.819 (linear), 0.805 (kNN) and 0.718 (nearest-class mean) in macro-F1 against C1's 0.884, 0.856 and 0.889; C1 is robust to the readout while C3 loses under all three, and although kNN beats the class-mean readout on C3 — so non-linear structure is genuinely present — it still does not reach C3's own linear probe. *Fine-tuning:* unfreezing everything and training with CE from the SupCon initialization (C3-ft) was pre-registered with the hypothesis "comparable to C1 within one point" and falsified it, settling at 0.818 on validation — its own linear-probe ceiling — and 0.720 on test; later epochs never beat its warm-up best. *Complementarity:* concatenated C1$\oplus$C3 features score 0.868 on validation, *below* the same-loss ensemble control C1$\oplus$C1′ (0.888): SupCon is a worse partner than a second CE encoder, though both probes stop early on the fragile validation split, so the direction is firmer than the magnitude. *Geometry:* the left column of Fig. 3 shows why a linear readout struggles — C3's low-motion classes (L, S and the empty room) chain into one filament instead of three separable blobs — and the same diagnostic on C3-ft (not shown) shows the fine-tune *partly* un-chaining C3's geometry, turning the SupCon encoder back into C1 rather than building on it. Counted honestly this is two families of evidence, not four instruments: the readouts share cached features and agree partly by construction, while the fine-tune is a different path and carries the confirmation — it is also the arm that controls the one difference the design cannot remove, since it retrains the encoder under C1's own sampler and augmentation profile. Both point the same way — the SupCon features are less informative about the activity than CE's, and no head, linear or not, recovers what they do not encode.

D. The adversary costs accuracy and stability

C2 loses 15.8 accuracy points on test. The direction is stable on both seeds (seed 43 against seed 43, 20.0 points) and on validation, but the interval $[-0.323, +0.034]$ crosses zero: the same 11 traces resolve half that gap for C3, so the limit is not their number but their agreement, C2's damage falling on the H, W and S traces. The same-configuration spread is 0.5 points for C1 against 4.7 for C2: the adversary appears to cost stability as well as accuracy. A linear probe on C2's *frozen* features reaches 0.720 test accuracy, 7 points above its own trained head (0.647), against $+0.002$ for C1: much of the damage sits in the head, not the representation.

E. Why: the adversary has no target

We answer the natural question — why an invariance objective fails on a transfer benchmark — by probing the *frozen* encoders along the dashed path of Fig. 2. On a trace-disjoint inner split of the training traces we fit linear probes for the activity (positive control) and for every domain attribute, each against its own majority baseline. The control saturates (1.000 against a 0.197 baseline) while *every* domain attribute sits at or below its own: capture set -0.090 against a 0.286 baseline — the non-degenerate target, and the one the adversary optimizes — direct path -0.098 and monitor position -0.086; the environment and subject probes collapse to the constant majority predictor, though at baselines of 0.854 and 0.818 those two are weak tests, degenerate for the same incidence reason that carries the argument below. None of them is linearly readable from the features, on precisely the data an in-training adversary reads. The same probes on the C2, C3 and C3-ft encoders, on all three contrastive checkpoints and on the second rotation — five encoders, C2 by construction — return the same verdict, the largest positive deviation anywhere being $+0.015$; the right column of Fig. 3 is the geometric counterpart, capture-set colours uniformly mixed inside every activity cluster.

A GRL removes only what its discriminator reads — and our probe is linear, but the adversary is an MLP; here there is nothing to read, and the adversary head shows it directly — on both seeds it never left the majority-class floor, already at epochs 1–2 with λ still ramping, so it never became a discriminator at all. What reaches the encoder is then gradient noise, and Section VI-D locates its damage: the representation

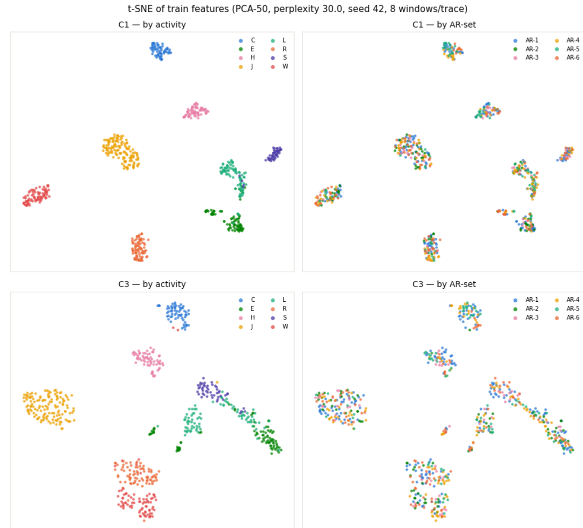

Fig. 3: t-SNE [10] of train features; top row C1 (CE), bottom row C3 (SupCon). *Left, by activity:* both form tight clusters, but C3's L/S/E clusters chain into one filament instead of three separable blobs — the geometric counterpart of its lower linear-probe score. *Right, the same points by capture set:* colours are uniformly mixed inside every cluster, in both encoders — no domain sub-structure for an adversary to remove.

survives it, the jointly trained head does not. The implication is structural: with the bedroom in five capture sets and one set for each other room (Section IV), any rotation trains either on a single environment or on one environment plus a single session, so environment invariance is barely definable on this training distribution, and C4 was closed untrained on that basis — evidence-based triage, not a claim that the GRL is useless in general. The probe alone shows only that the domain is not *linearly* decodable from traces the encoder has seen; the structural argument is what generalizes the verdict.

F. Error structure and limitations

Checkpoint selection was blind to C, E and S, and we had pre-declared the risk that those classes would underperform; the opposite holds, the val-blind classes being the *best* (0.904 mean F1 against 0.744), so class difficulty in the target environment, not selection blindness, drives per-class error. The errors themselves are physically sensible (Fig. 4): C1 confuses mainly walking$\leftrightarrow$running and H$\rightarrow$J, pairs with adjacent Doppler signatures, and recognizes the empty room perfectly; its largest remaining confusion is S$\rightarrow$L, a pair inside the low-motion group that C3's embedding chains into one filament. The confusions are between physically similar activities, i.e. genuine motion features rather than room artifacts. Calibration dissociates from accuracy: every stream is *under*confident, C1 most (ECE 0.243; antenna averaging, plus label smoothing) while the GRL model is the best calibrated (0.070) and the least accurate objective, so any deployment consuming confidence scores needs temperature scaling.

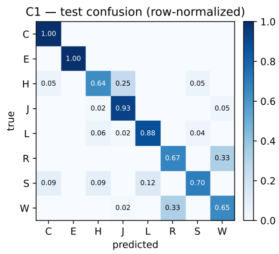

Fig. 4: Confusion matrix of C1 on the held-out laboratory session (row-normalized).

Scope and limitations. The same session also evaluated two further pre-registered arms that we do not discuss for lack of space — test-time adaptation (AdaBN [11], T3A [12]) and amplitude augmentation — neither of which resolved an improvement over C1 (best $+0.073$, on the second rotation, interval crossing zero). Beyond those, model selection is in-domain (9 held-out training traces), so the selected checkpoints may be sub-optimal out of domain; each held-out set is a single session, so “environment” is confounded with subject, monitor position and day; and at 11 and 15 traces a percentile bootstrap separates two or three performance tiers, not a dozen individually ranked rows.

VII. Concluding Remarks

We evaluated, under one shared apparatus and a single-use test protocol, whether adversarial (GRL) and contrastive (SupCon) objectives improve cross-environment WiFi CSI activity recognition over plain cross-entropy. They do not: SupCon is worse by a bootstrap-resolved margin and the GRL is worse still and less stable across seeds, while the deep CE model beats the shallow–wide reference design by 23.5 accuracy points — depth mattered more here than any invariance objective. What we consider most useful are the two reasons *why*. The contrastive encoder fails in the representation itself: kNN and nearest-class-mean readouts, a concatenation control and a full CE fine-tune all decline to recover the gap, so its features carry less of what tells the activities apart. The adversary fails for the complementary reason: frozen-encoder probes show that no domain attribute is linearly readable from the learned features, so it has nothing to remove. Both preconditions cost one probe on cached features to check, and both are worth checking — on any dataset whose training split, like this one, is dominated by a single environment — a message that goes to benchmark designers as much as to practitioners.

A. Challenges and Lessons Learned

Much of this project lay outside the material we had seen: test-time adaptation, the trace-level paired bootstrap and the frozen-encoder probing protocol were designed from the literature rather than assembled from known recipes, and while contrastive learning was familiar in its self-supervised form, its supervised variant and the gradient reversal layer had to be implemented from the original papers. The hardest decisions came from the data, imbalanced along every axis at once — environments, capture sets, subjects and classes — so that each remedy costs something: pinning rare cells into training guarantees coverage but empties part of validation, and balancing contrastive batches over capture sets oversamples trace-poor sets. We also learned to distrust single numbers: the seed replicates showed a same-configuration swing on the GRL larger than some cross-configuration gaps, and a 16-point gap stayed unresolved where an 8-point one did not — across traces, agreement counts as much as size. Finally, a caveat we had declared in advance — that classes invisible to validation would suffer — proved false on test: plausible methodological worries should be measured, not just declared.

Future work. With one held-out session per rotation this is a transfer study rather than full domain generalization; the natural extension is a dataset with several well-populated training environments, where our diagnostic protocol transfers unchanged. Whether *any* training-time objective can help when the domain factor is unreadable remains open — candidates acting on the input, such as physics-informed augmentation, are unaffected by our structural argument.

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